PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

Title of your Project

Airbus ship detection

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Project Guide

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Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement
- System Design
- Technologies and methodologies
- Implementation
- Conclusion
- References

Sustainable Development Goals (SDG) mapped

SDG 13 – Climate Action

- Satellite-based detection helps track:
  - Marine pollution
  - Environmental impact
- Supports climate monitoring efforts

SDG 16 – Peace, Justice and Strong Institutions

- Helps in:
  - Maritime surveillance
  - Detecting illegal activities (smuggling, piracy)
- Improves **security and governance**

Introduction

Monitoring ship activity in oceans is important for maritime safety and security. Traditional monitoring methods are time-consuming and cannot cover large sea areas effectively. Airbus Ship Detection uses satellite images and machine learning techniques to automatically detect ships. This helps improve maritime surveillance, monitor shipping traffic, and support research in computer vision and image detection.

Literature Survey of the existing system

Authors	Title	Year	Publisher/source	Methodology used
Al-Saad, M., Aburaed, N., Panthakkan, A., Al Mansoori, S., Ahmad, H. A., & Marshall, S.	Airbus Ship Detection from Satellite Imagery using Frequency Domain Learning	2021	Proceedings of SPIE (Image and Signal Processing for Remote Sensing XXVII)	Used deep learning approach with an enhanced Faster R-CNN (FRCNN) model that combines high-level and low-level features. Applied frequency domain learning using Discrete Wavelet Transform and trained on the Airbus Ship Detection dataset from Kaggle.
Ronneberger, O., Fischer, P., & Brox, T	U-Net: Convolutional Networks for Biomedical Image Segmentatio	2015	MCCAI(springer)	Proposed U-Net CNN architecture for image segmentation. Uses encoder-decoder structure with skip connections. Widely adapted for satellite ship detection by segmenting ships from background pixels.

Authors	Title	Year	Publisher/source	Methodology used
Joseph Redmon et al.	You Only Look Once: Unified, Real-Time Object Detection	2016	CVPR	Single-stage object detection model dividing image into grids. Predicts bounding boxes and class probabilities in one pass. Applied to ship detection for real-time performance.

Limitations of existing systems

Weather Dependency Optical satellite images cannot penetrate clouds, fog, or heavy rain. These weather conditions can obstruct the view of the sea surface, preventing ships from being detected.Small Vessel Detection IssuesVery small boats or fishing vessels may not be detected because they are smaller than the spatial resolution of the satellite imagery.False PositivesNatural ocean features such as waves, wakes, offshore structures, or floating debris can sometimes be incorrectly identified as ships by the detection system.Occlusion and ClutterShips located near coastlines, ports, or close to other vessels may be difficult to distinguish, which can reduce detection accuracy.

Problem statement

The increasing number of ships operating across oceans makes manual monitoring of maritime traffic difficult and inefficient. Traditional surveillance methods require significant human effort and are unable to cover large ocean areas effectively. Therefore, there is a need for an automated system that can detect and identify ships from satellite images accurately. Using the Airbus Ship Detection dataset, machine learning and computer vision techniques can be applied to automatically detect ships in satellite imagery, helping improve maritime surveillance, security, and monitoring of sea activities.

System Design

SATELLITE IMAGERY INPUT → PREPROCESSING & AUGMENTATION → PREPROCESSING & AUGMENTATION → CNN/DENSE-NET & DETECTION

ML MODEL & ANALYSIS

- CNN Architecture (DenseNet-169)
- (Random Forest Comparison - Optional)
- Backpropagation Training
- Key Loss Metrics (e.g., BCE Loss)
- Adam Optimizer configuration

INPUT SOURCES

- Satellite Images (e.g., Airbus dataset)
- Location Data (Encoded masks in RLE)
- Image Metadata (e.g., Sea conditions)
- Resolution parameters
-

PROCESSING PIPELINE

- Data Cleaning & Standardization
- RLE Decoding into Binary Masks
- Normalization (e.g., [0,1] range)
- Feature Engineering (Preprocessing)
- Data Splitting (Train / Val / Test)
- Image Augmentation (Rotation, Flipping, Noise)
- Loss Function Definition (e.g., Binary Cross Entropy)

ML MODEL & OUTPUT

CNN

- CNN Model (e.g., DenseNet-169)
- Feature Extraction Layers (Edges, Textures)
- (Optionally Random Forest Baseline)
- Object Detection Techniques
- Predicted Ship Locations (Bounding Boxes/Masks)
- Model Optimization (e.g., Adam Optimizer)

INPUT SOURCES → Processing → Feature Extraction → CNN Training → Object Detection → OUTPUT RESULTS

PERFORMANCE EVALUATION (Objective 4)

- IoU (Intersection over Union)
- Precision Score
- Recall Score
- F1-Score
- Confusion Matrix (Ship vs No-Ship)

Technologies and methodologies

Technologies:

- Backend: Python (Fast api for handling model logic, inference requests, and running locally)
- Frontend: react with next.s(user interface for image upload, prediction results, comparison slider, and dashboard)
- Language:Python 3.11, javascript(ES6+), Typescript

Libraries and tools:

- NumPy and pandas(data preprocessing)
- Scikit-learn(traditional ml models)
- Tensorflow(deep learning models)
- Opencv, scikit-image(image preprocessing)
- Matplotlib(visualisation)

Technologies and methodologies

Dataset:

- Dataset Name: Airbus ship detection dataset
- Dataset type: satellite imagery dataset
- Dataset size: 190000+ images

Technologies and methodologies

Methodologies:

Image Input System →

User uploads satellite images through the frontend interface. Images are preprocessed (resizing, normalization) before being passed to models.

Prediction System →

Uses multiple machine learning approaches to detect ships:

Binary classification output (Ship / No Ship)

Probability-based prediction scores

Technologies and methodologies

Model approaches:

1. Traditional Machine Learning

Uses handcrafted features:

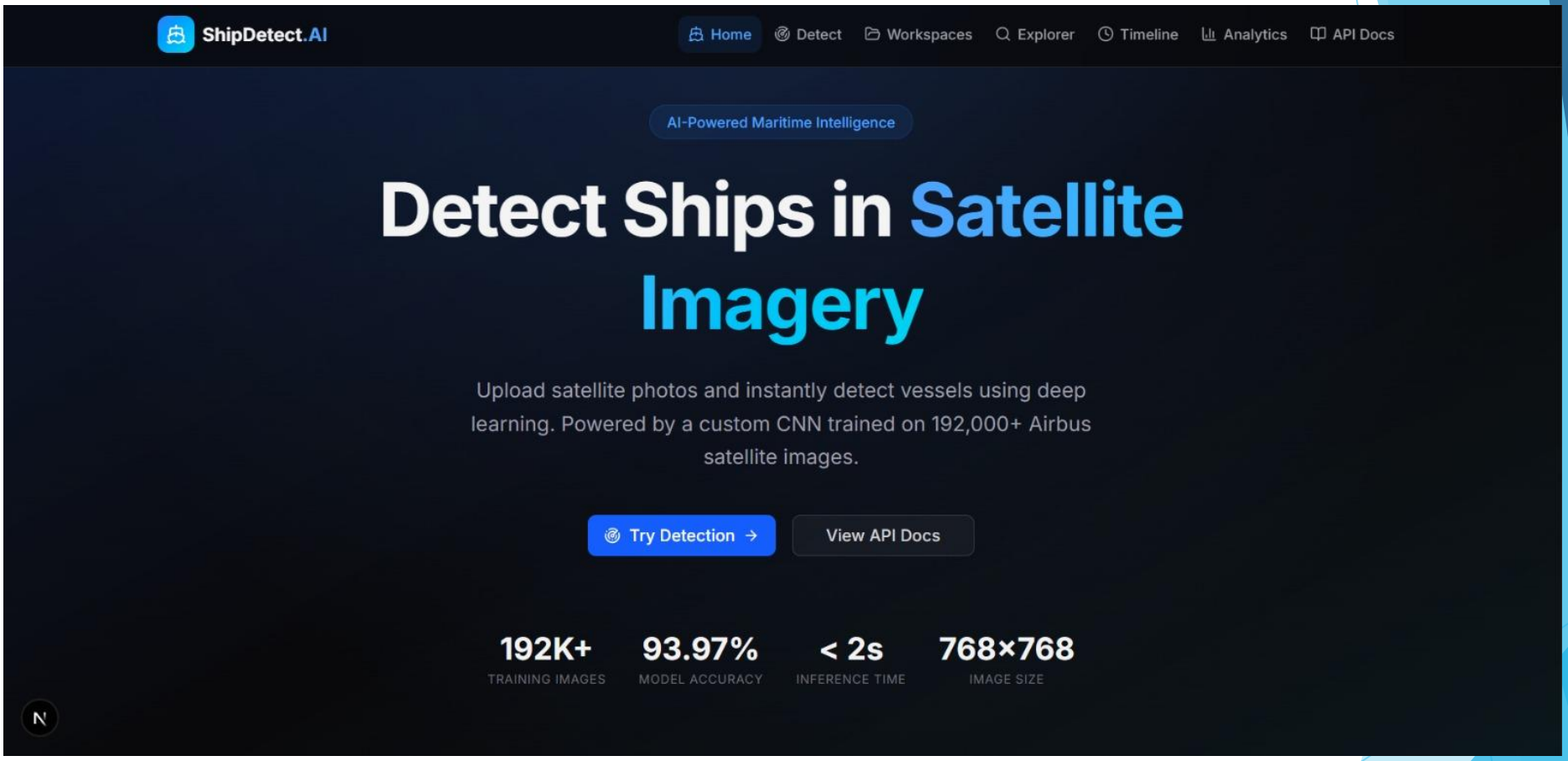
- Hu Moments (shape)
- Haralick features (texture)
- Color histograms

Models: LDA, KNN, SVM, Naive Bayes, Random Forest

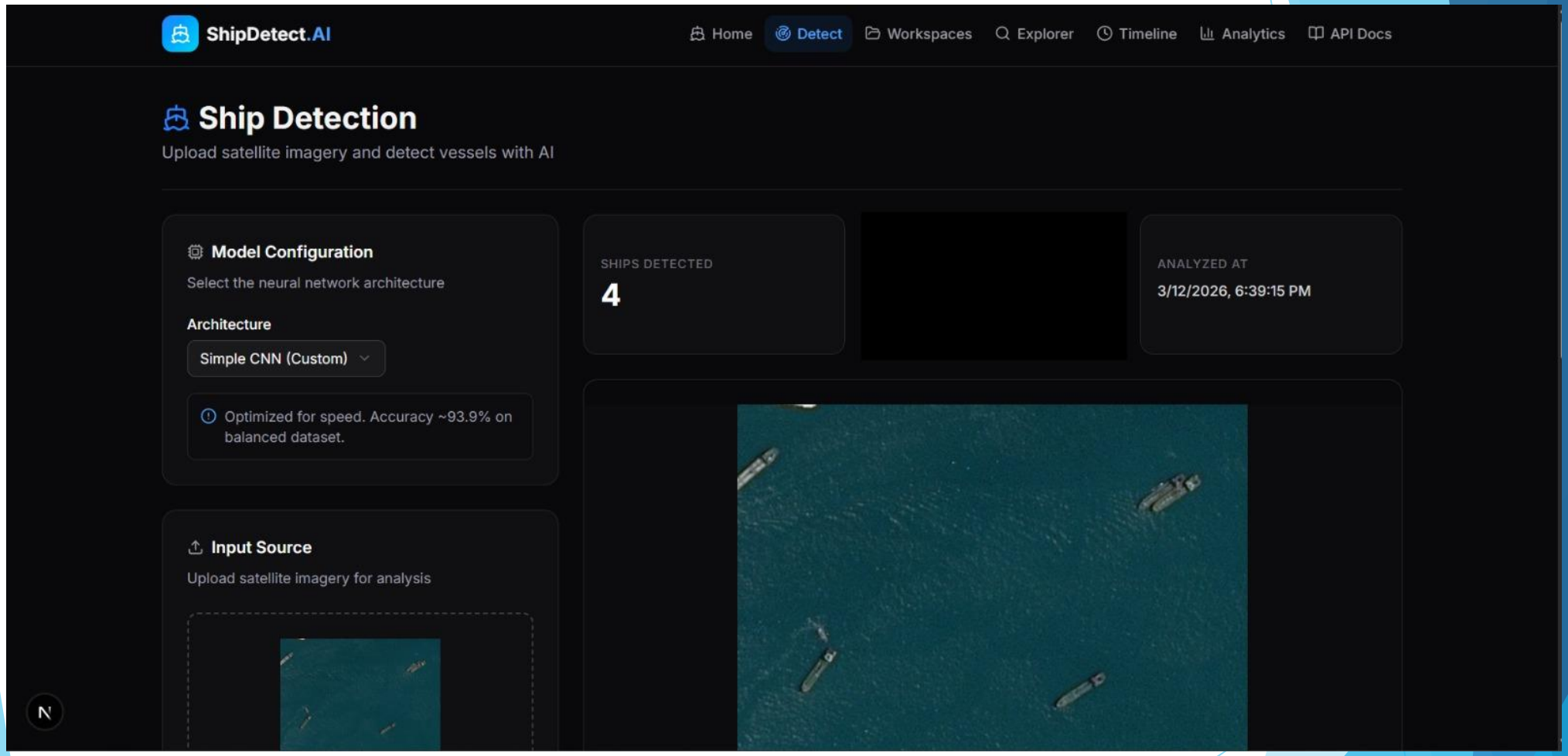
Custom CNN Model

- Built using PyTorch
- 3 convolution layers + dense layers
- Uses data augmentation (flip, rotate)
- Trained on balanced dataset

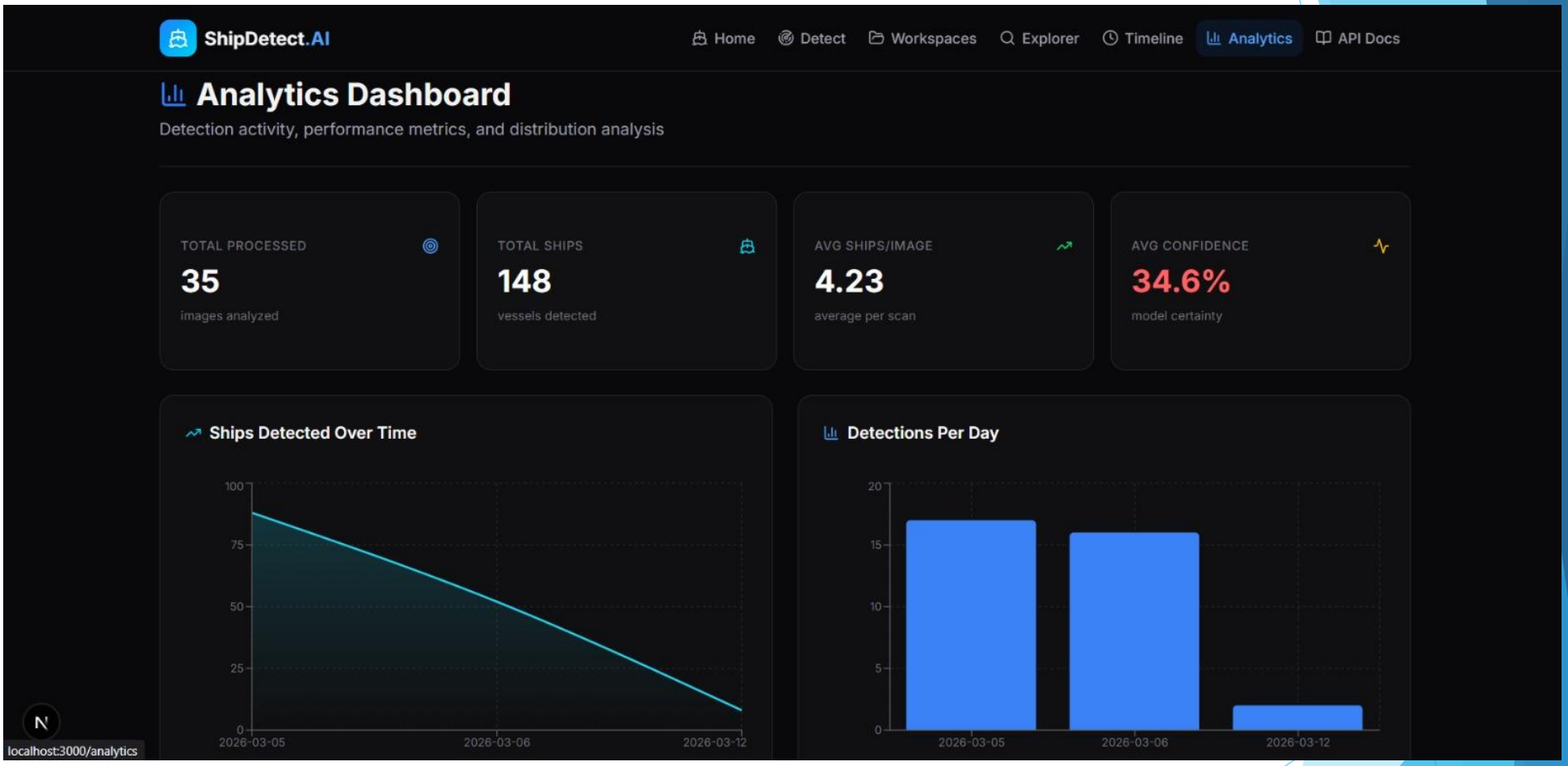
Implementation



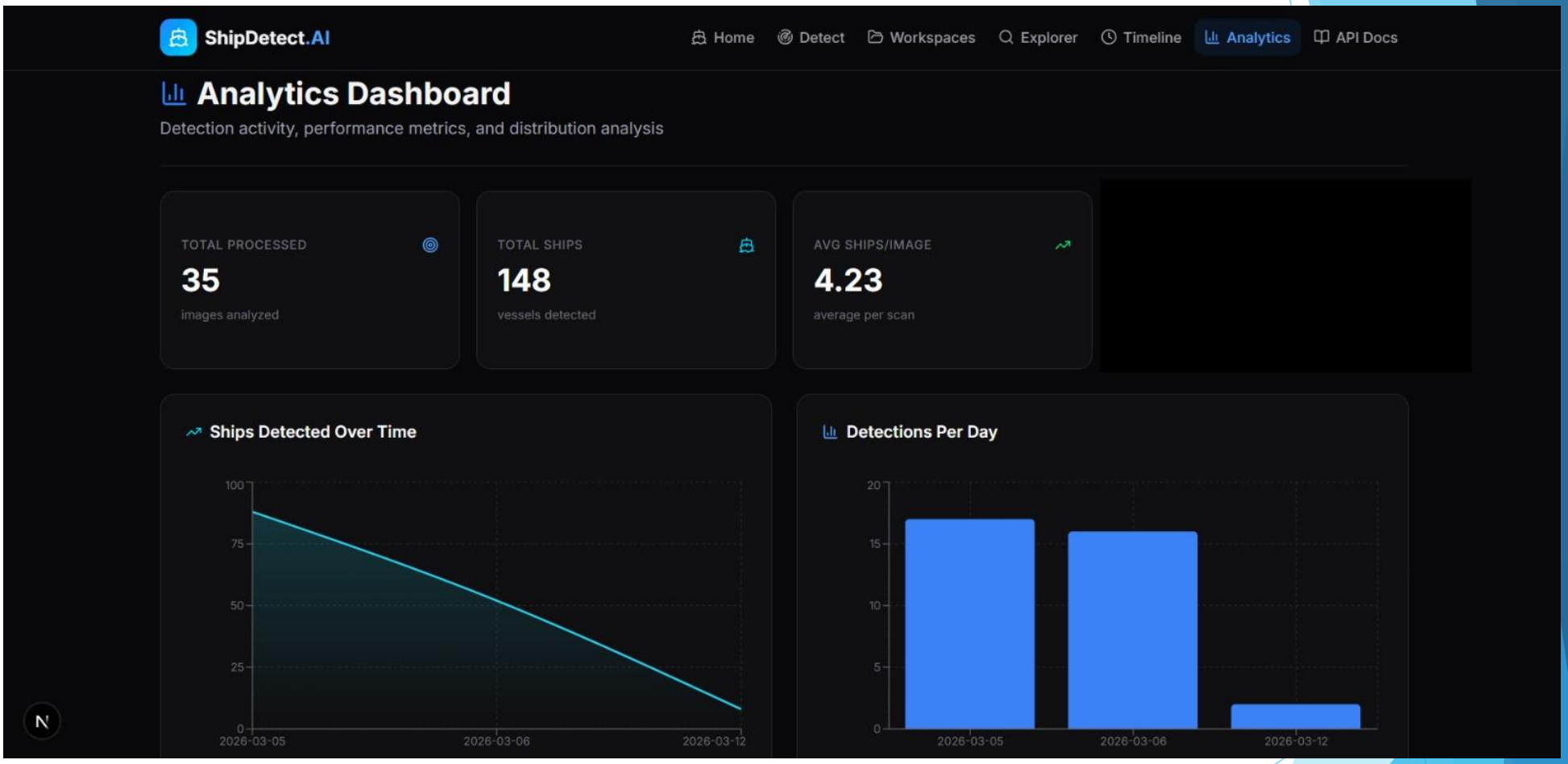
Implementation



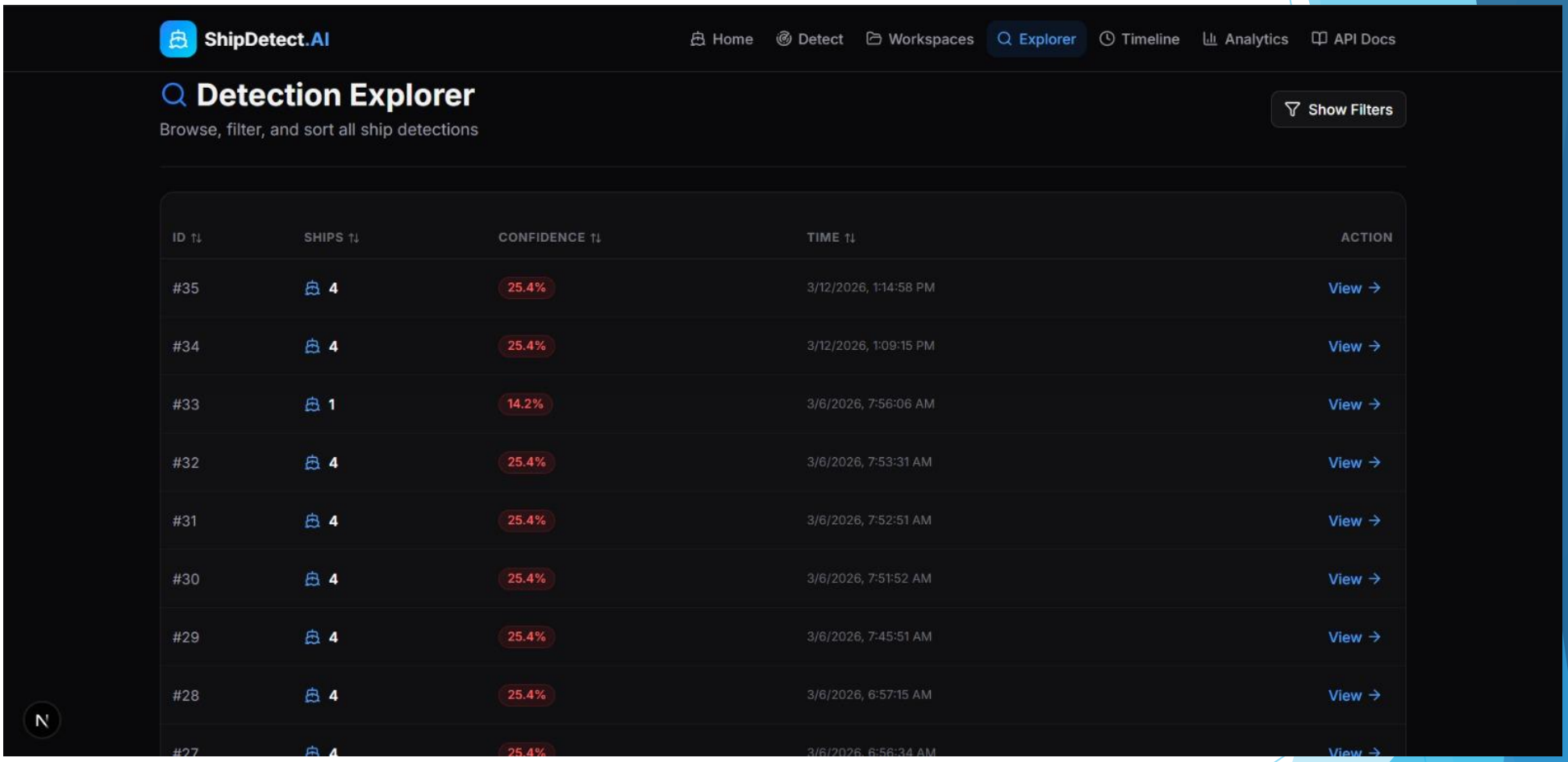
Implementation



Implementation



Implementation



Conclusion

Airbus Ship Detection using satellite imagery and machine learning provides an efficient way to automatically identify ships over large ocean areas. It helps improve maritime surveillance, monitoring of ship traffic, and detection of illegal activities at sea. By using advanced image processing and machine learning techniques, the system can analyze satellite images and detect ships more quickly and accurately than manual methods. Although there are some limitations such as weather conditions and detection of small vessels, this approach significantly improves the efficiency and effectiveness of maritime monitoring.

References

1. Al-Saad, M., Aburaed, N., Panthakkan, A., Al Mansoori, S., Ahmad, H. A., & Marshall, S. (2021). Airbus Ship Detection from Satellite Imagery using Frequency Domain Learning. Proceedings of SPIE – Image and Signal Processing for Remote Sensing XXVII, 11862, 118620Z. SPIE. https://strathprints.strath.ac.uk/79444/1/Al_Saad_etal_SPIE_2021_Airbus_ship_detection_from_satellite_imagery.pdf
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Thank You!!